



DATA ANALYSIS AND VISUALIZATION

INSTRUCTOR: UMME AMMARAH





IMAGE ENHANCEMENT

SPATIAL DOMAIN METHODS

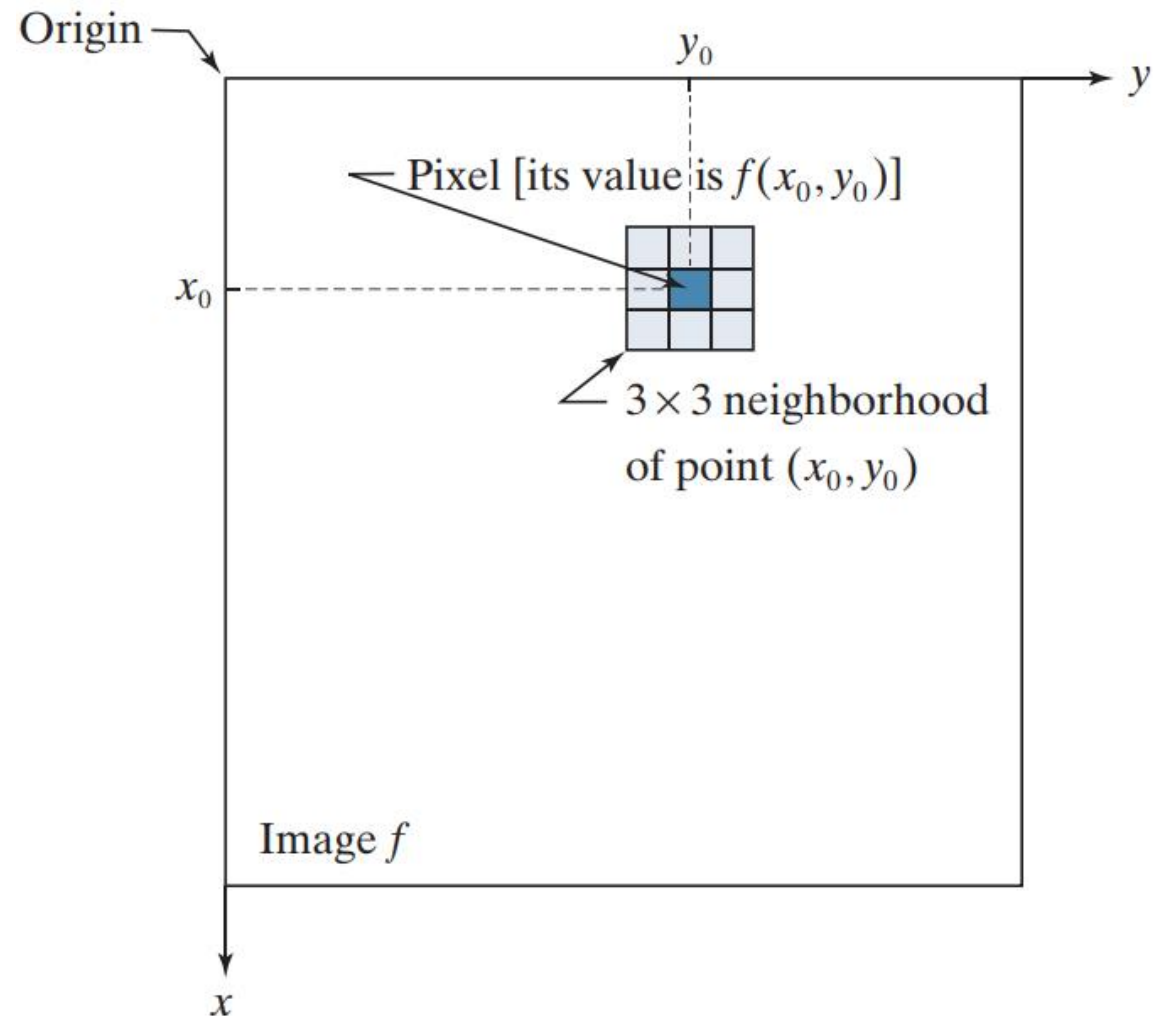


IMAGE ENHANCEMENT

- To process an image so that the result is more suitable than the original image for a specific application.
- Categories:
 - Spatial domain methods
 - Frequency domain methods

SPATIAL DOMAIN METHOD

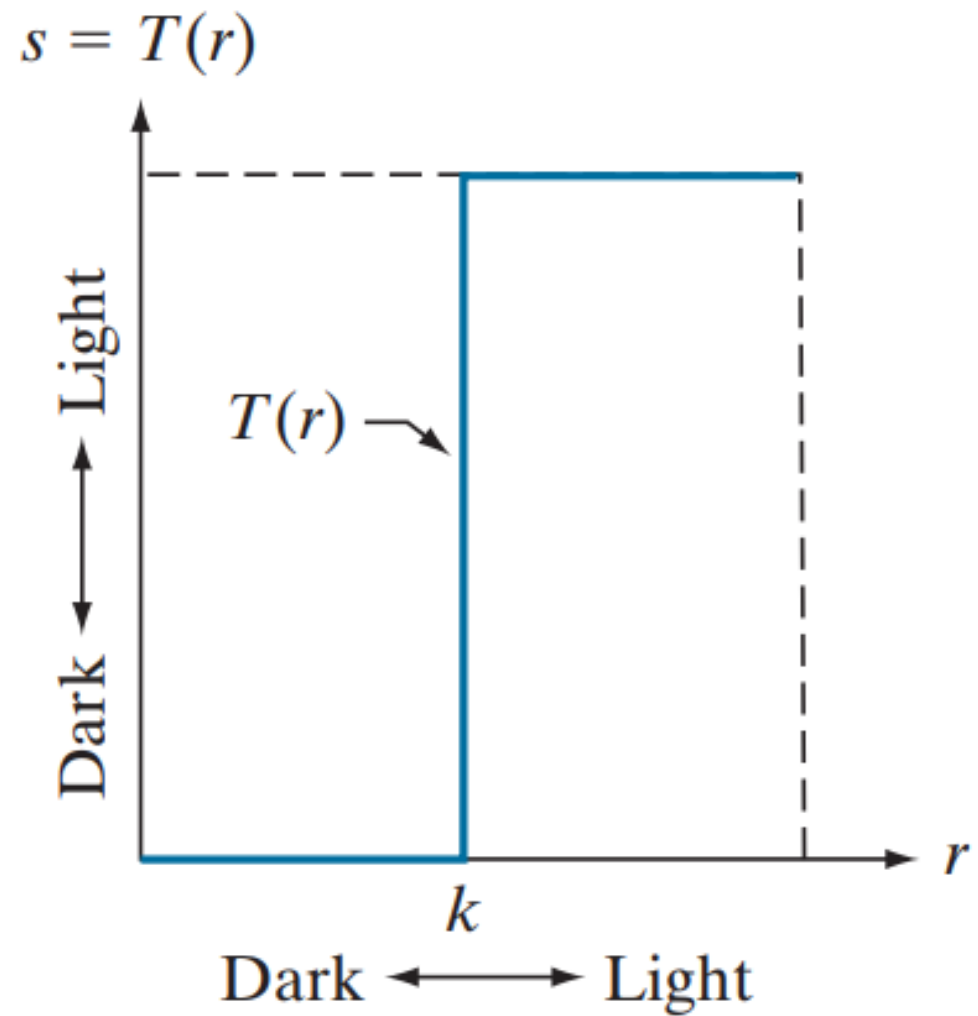
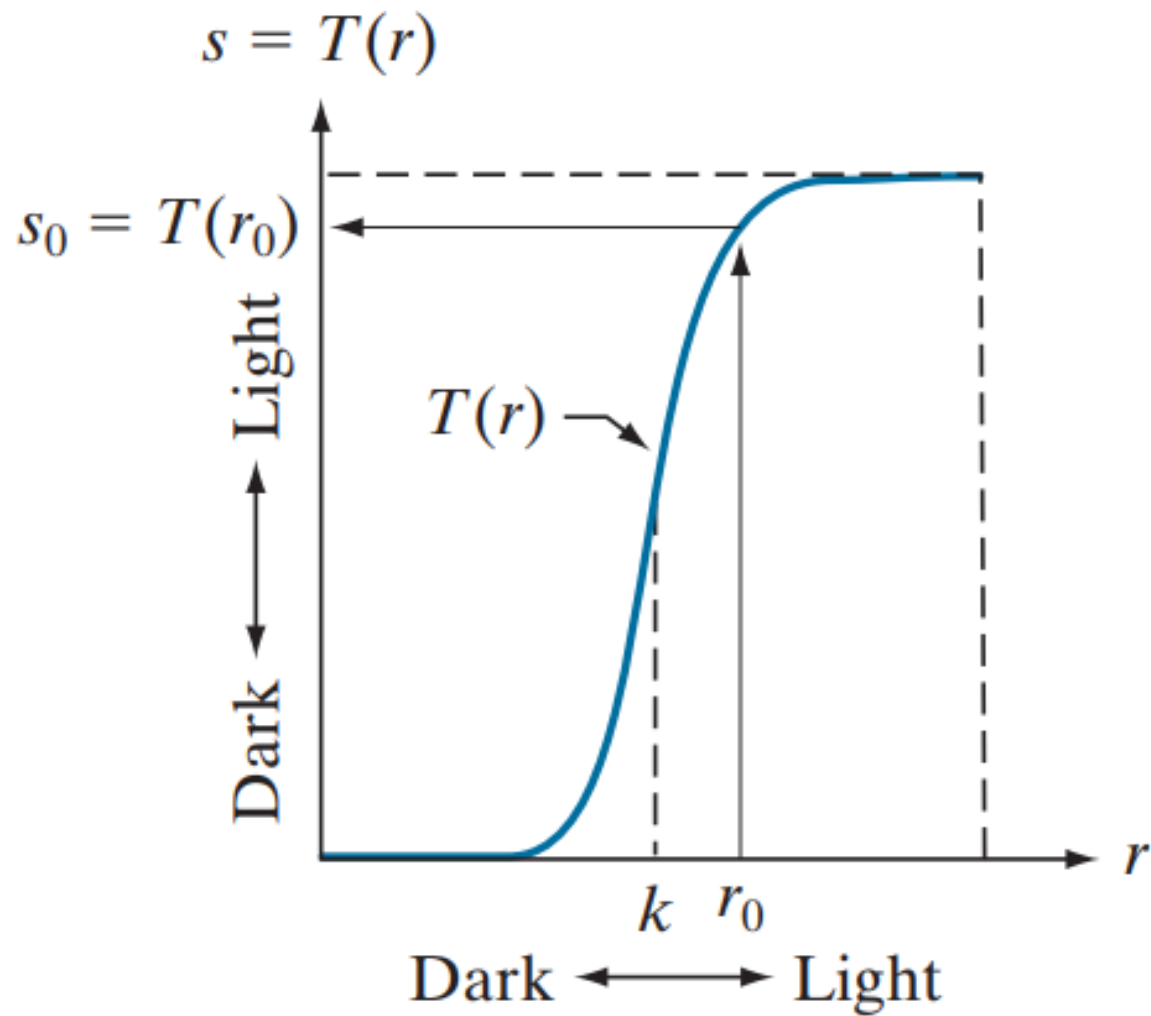
A 3×3 neighborhood about a point (x_0, y_0) in an image. The neighborhood is moved from pixel to pixel in the image to generate an output image.



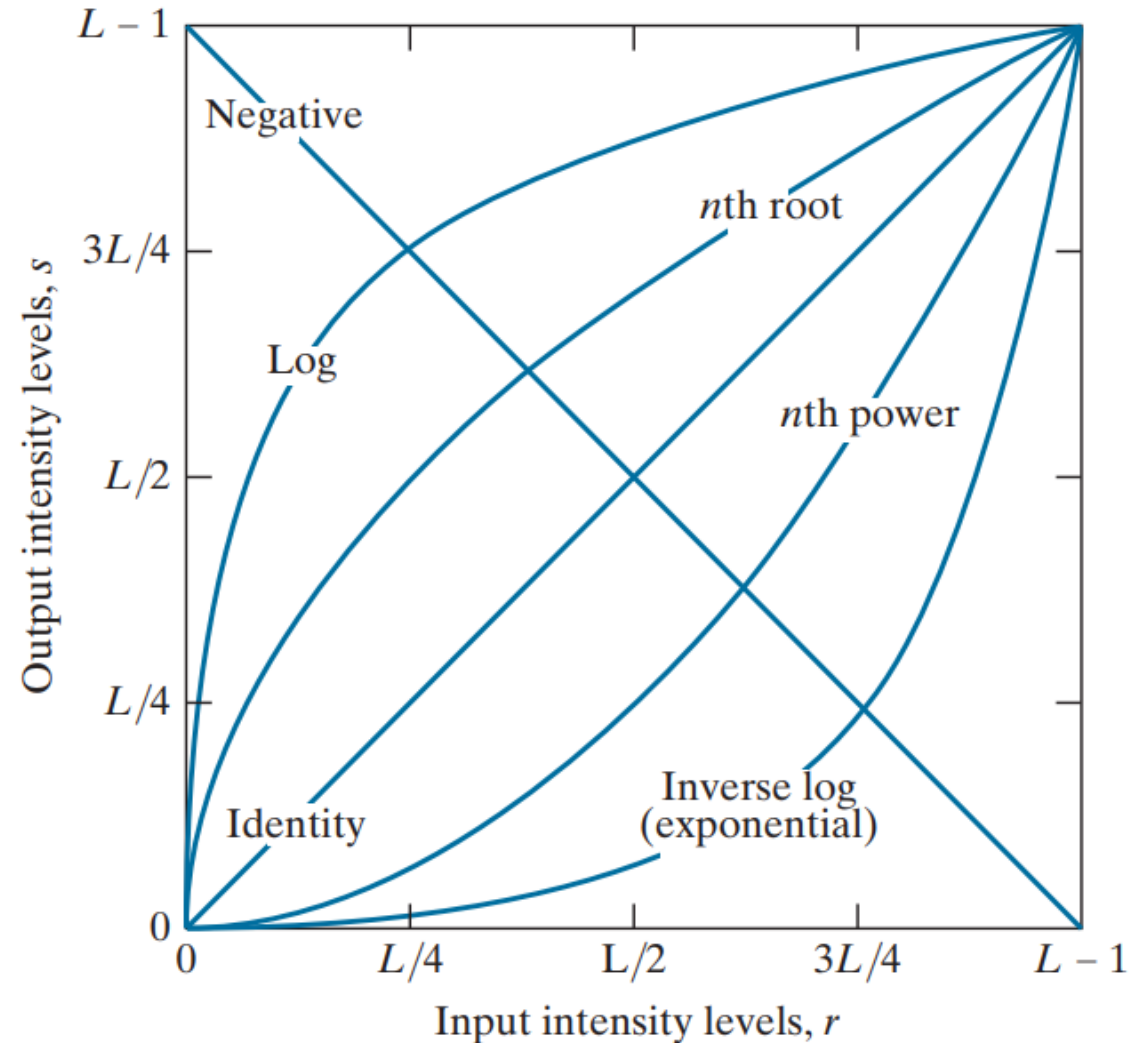
SPATIAL DOMAIN METHODS

- When the neighborhood is 1×1 then g depends only on the value of f at (x, y) and T becomes a gray-level transformation (or mapping) function.
- Examples: Point processing techniques (e.g. contrast stretching, thresholding)

CONTRAST STRETCHING AND THRESHOLDING



INTENSITY TRANSFORMATION



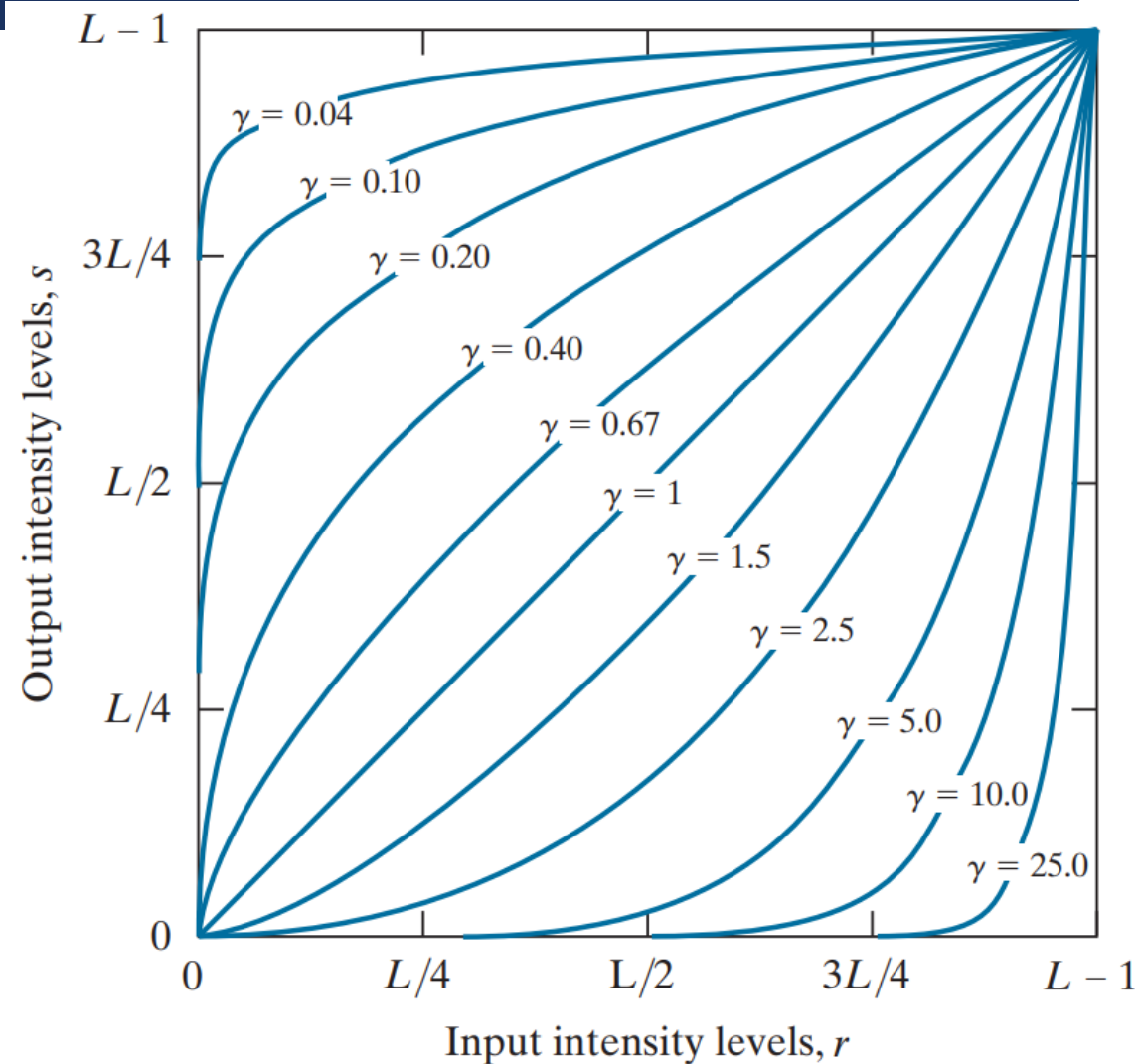
LOG TRANSFORMATIONS

- Log function: $s = c \log(1 + r)$
 - Stretch low intensity levels Compress high intensity levels
- Inverse log function: $s = c \log^{-1}(r)$
 - Stretch high intensity levels Compress low intensity levels

POWER LAW (GAMMA) TRANSFORMATION

- $S = cr^\gamma$

- Gamma correction/gamma encoding



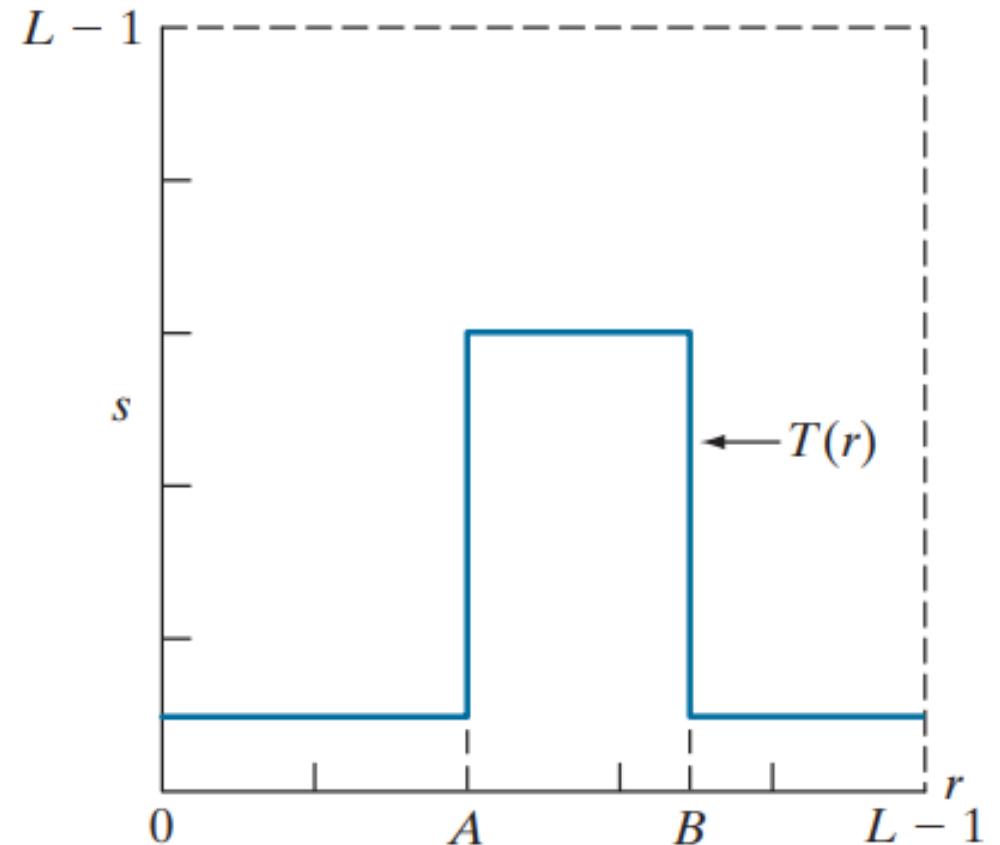
Results of applying the transformation with $g = 3.0, 4.0, \text{ and } 5.0$, respectively.
($c = 1$ in all cases.)

Washed-out appearance was reduced by a large gamma value



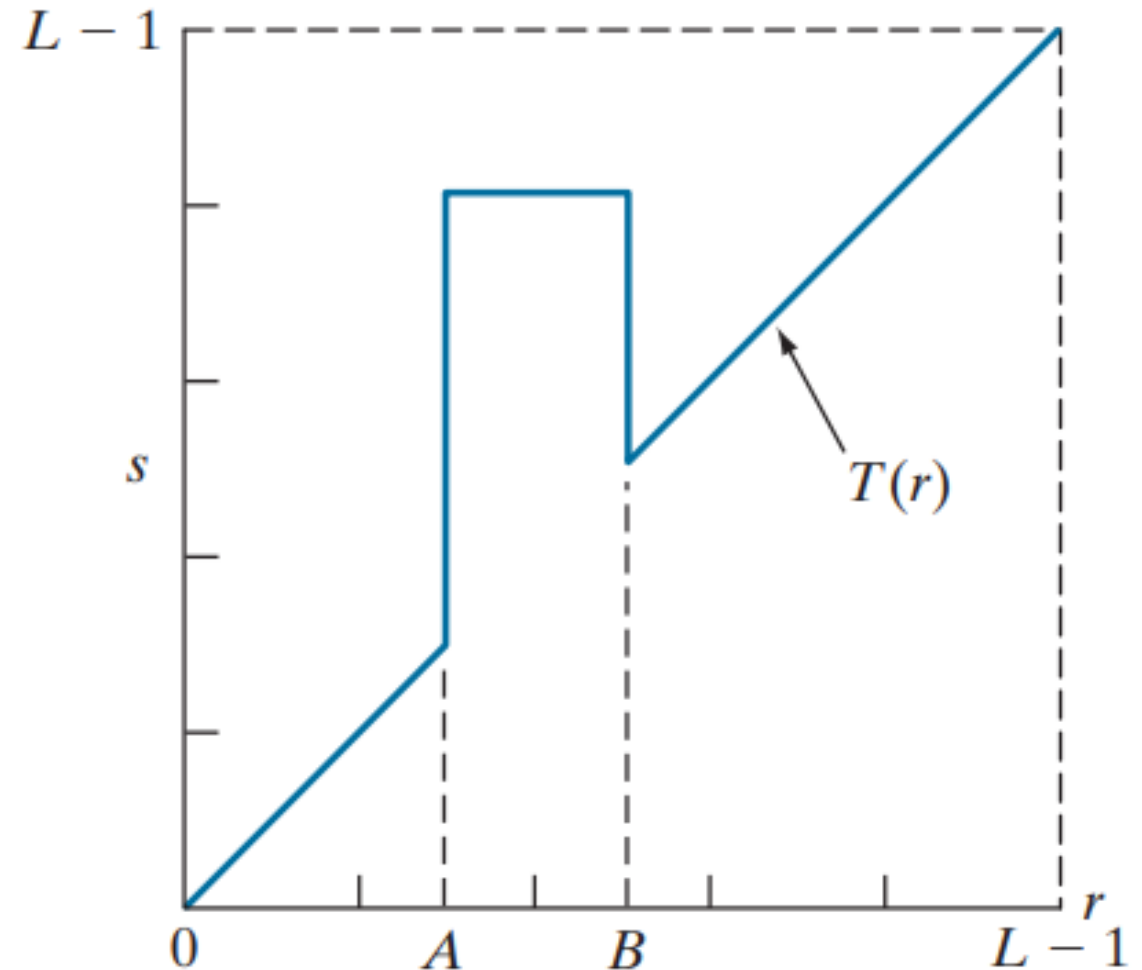
INTENSITY LEVEL SLICING

- To highlight a specific range of gray levels in an image (e.g. to enhance certain features).
- One way is to display a high value for all gray levels in the range of interest and a low value for all other gray levels (binary image).



INTENSITY LEVEL SLICING

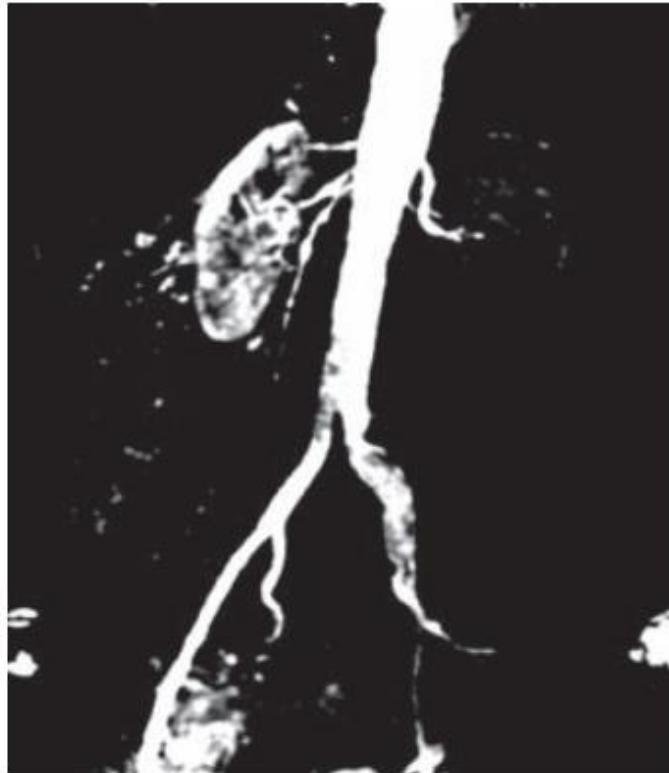
- The second approach is to brighten the desired range of gray levels but preserve the background and gray-level tonalities in the image.



EXAMPLE



Original Image



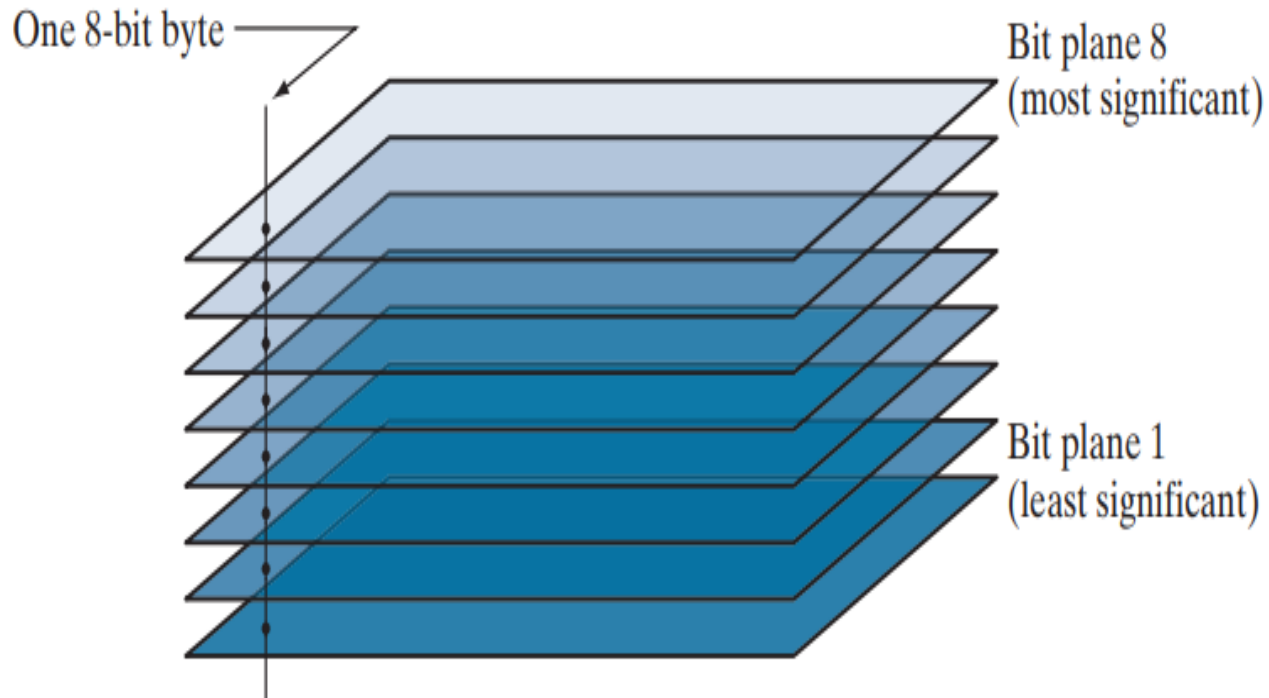
Result of using slicing method 1



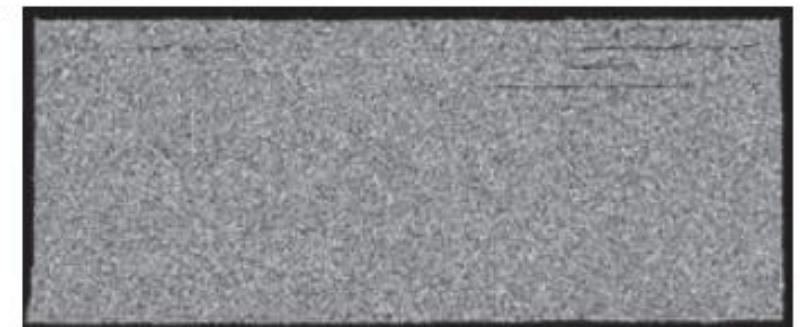
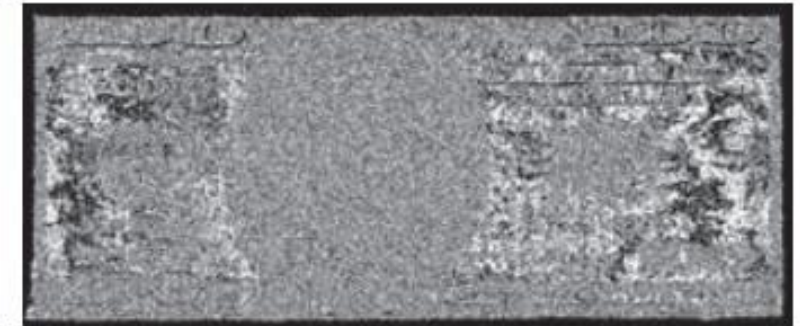
Result of using slicing method 2

BIT PLANE SLICING

- To highlight the contribution made to the total image appearance by specific bits.
 - Plane 1 contains the least significant bit and plane 8 contains the most significant bit.
 - Only the higher order bits (top four) contain visually significant data. The other bit planes contribute the more subtle details.
 - Plane 7 corresponds exactly with an image thresholded at gray level 128.



Each pixel is represented by 8 bits, the image is composed of 8 1-bit planes



a	b	c
d	e	f
g	h	i

(a) An 8-bit gray-scale image of size 550 x 1192 pixels. (b) through (i) Bit planes 8 through 1, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image..

IMAGE RECONSTRUCTED FROM BIT PLANES



(a) 8 and 7



(b) 8, 7, and 6



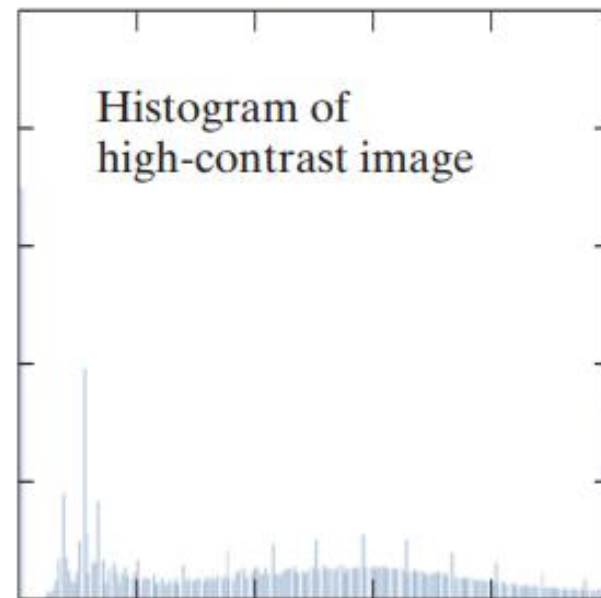
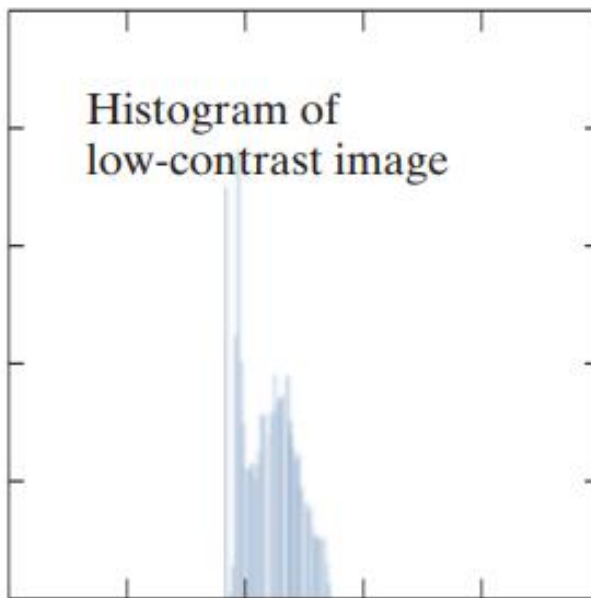
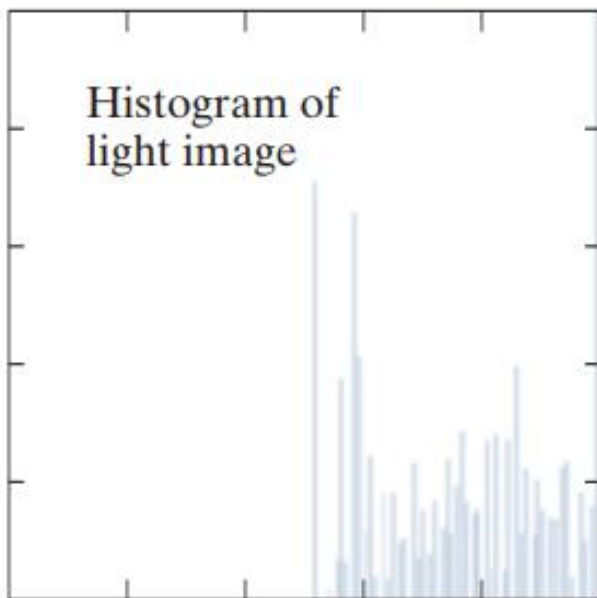
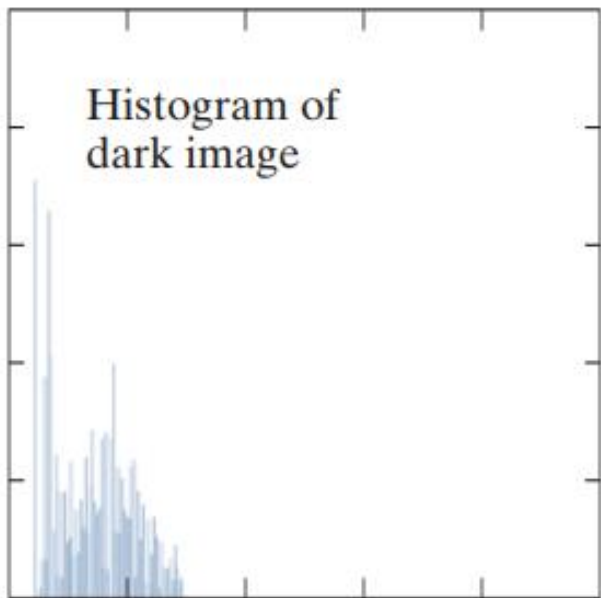
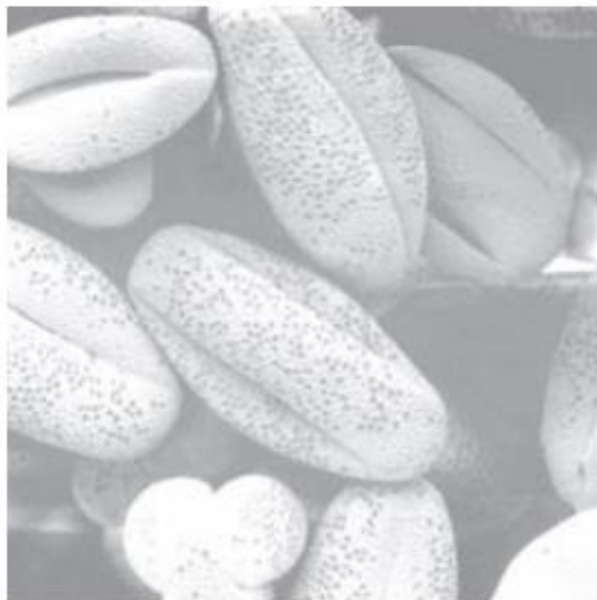
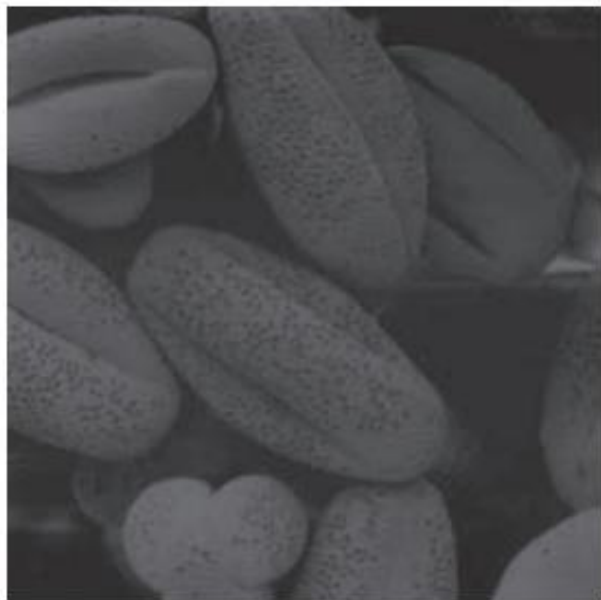
(c) 8, 7, 6, and 5

We conclude that, in this example, storing the four highest-order bit planes would allow us to reconstruct the original image in acceptable detail. Storing these four planes instead of the original image requires 50% less storage.



HISTOGRAM PROCESSING





METHODS OF HISTOGRAM PROCESSING

- Two well-known methods of image enhancement:
 - Histogram equalization
 - Histogram matching

IMAGE HISTOGRAM

- Mathematically represented as:

$$h(r_k) = n_k$$

- r_k represents the intensities of an L-level digital image. $k = 0, 1, 2, \dots, L-1$ and n_k is the number of pixels in f with intensity r_k

- Normalized Histogram:

$$p(r_k) = \frac{h(r_k)}{MN} = \frac{n_k}{MN}$$

- M and N are the number of image rows and columns, sum of $p(r_k)$ for all values of k is always 1.
- It estimates of the probabilities of intensity levels occurring in an image

IMAGE HISTOGRAM

- An image histogram is a plot of the gray level frequencies.

0	0	1	0	2	0
1	0	7	7	7	0
0	7	0	0	7	0
1	0	0	7	2	0
0	0	7	1	0	1
1	0	7	7	7	0

frequencies

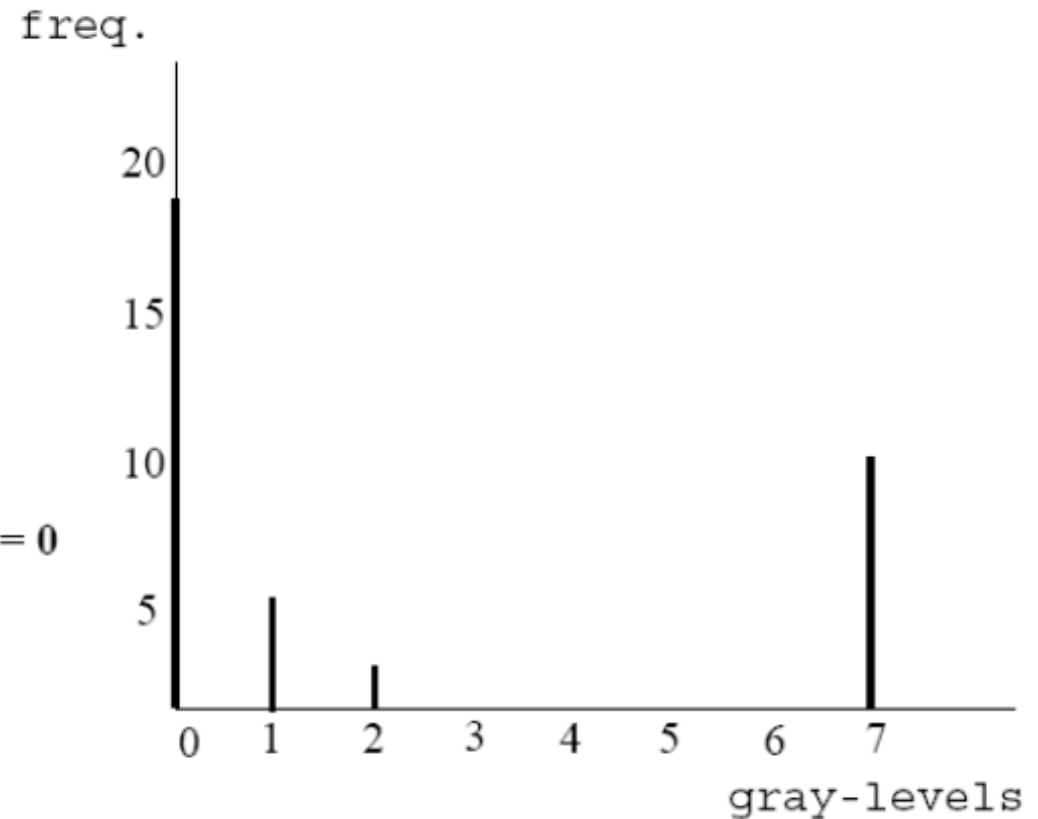
$$f(0) = 18$$

$$f(1) = 6$$

$$f(2) = 2$$

$$f(3) = f(4) = f(5) = f(6) = 0$$

$$f(7) = 10$$



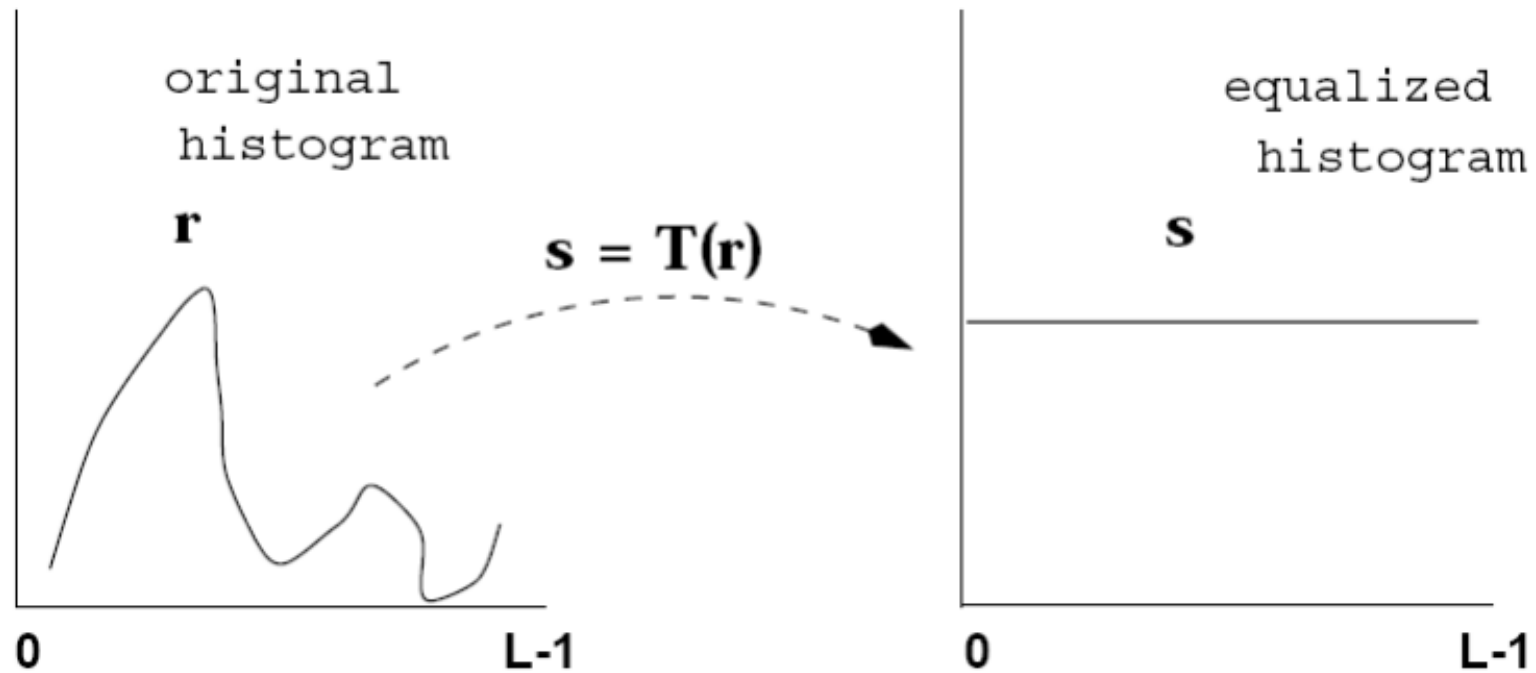
NORMALIZED HISTOGRAM

- Divide frequencies by total number of pixels to represent as probabilities.

$$\begin{aligned} P(0) &= \frac{f(0)}{36} = \frac{1}{2} & P(1) &= \frac{f(1)}{36} = \frac{1}{6} \\ P(2) &= \frac{f(2)}{36} = \frac{1}{18} & P(3) &= P(4) = P(5) = P(6) = 0 \\ P(7) &= \frac{f(7)}{36} = \frac{5}{18} \end{aligned}$$

HISTOGRAM EQUALIZATION

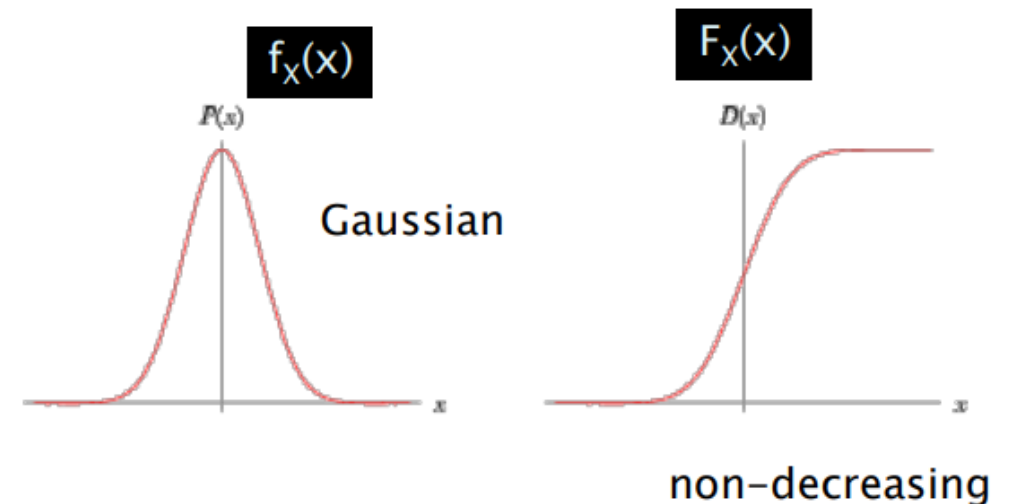
- The main idea is to redistribute the gray-level values uniformly.



PROBABILITY DENSITY/ PROBABILITY DISTRIBUTION

- The **probability density function (pdf)** is a real-valued function $f_x(x)$ describing the density of probability at each point in the sample space.
- The integral of $f_x(x)$ defines the **cumulative probability distribution function(cdf) $F_x(x)$**

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(a) da$$



EXAMPLE

In the discrete case:

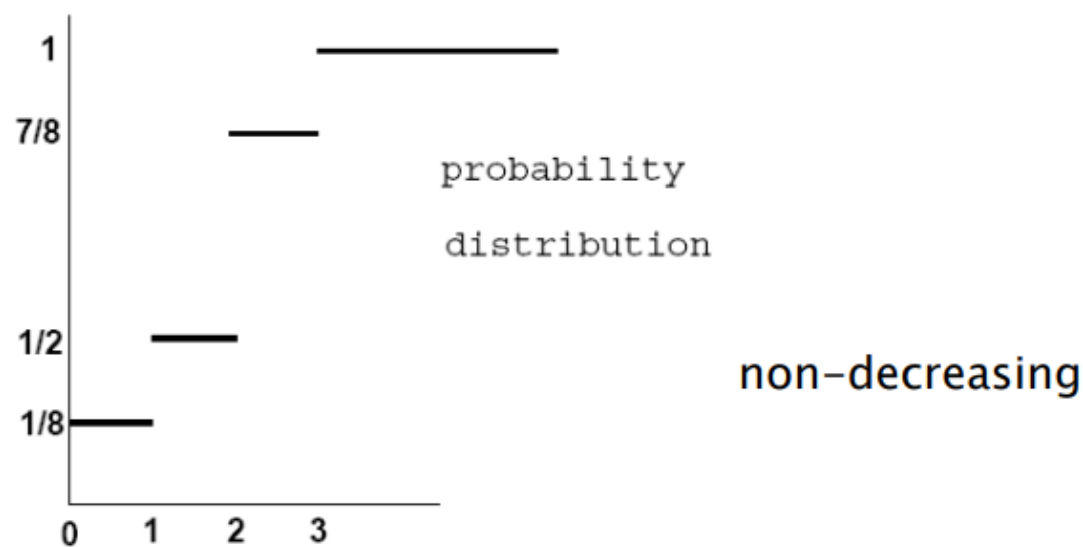
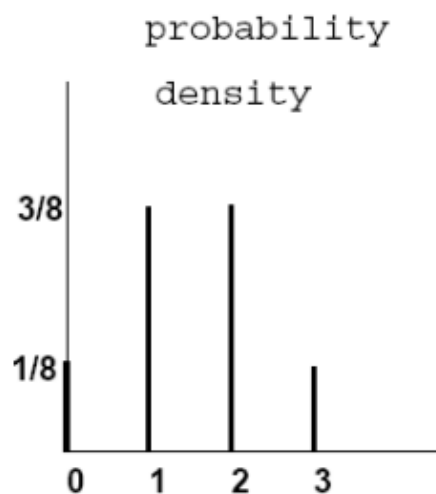
$$F_X(x) = P(X \leq x) = \sum_{k=0}^x P(X = k)$$

$$F_X(0) = P(X \leq 0) = P(X = 0) = 1/8$$

$$F_X(1) = P(X \leq 1) = P(X = 0) + P(X = 1) = 1/2$$

$$F_X(2) = P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = 7/8$$

$$F_X(3) = P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) = 1$$



EXAMPLE

1	2	1	1	1
2	5	3	5	2
2	5	5	5	2
2	5	3	5	2
1	1	1	2	1

Max Value = 5 (requires 3 bits)
 Number of grey levels (L) = 8
 N = 25

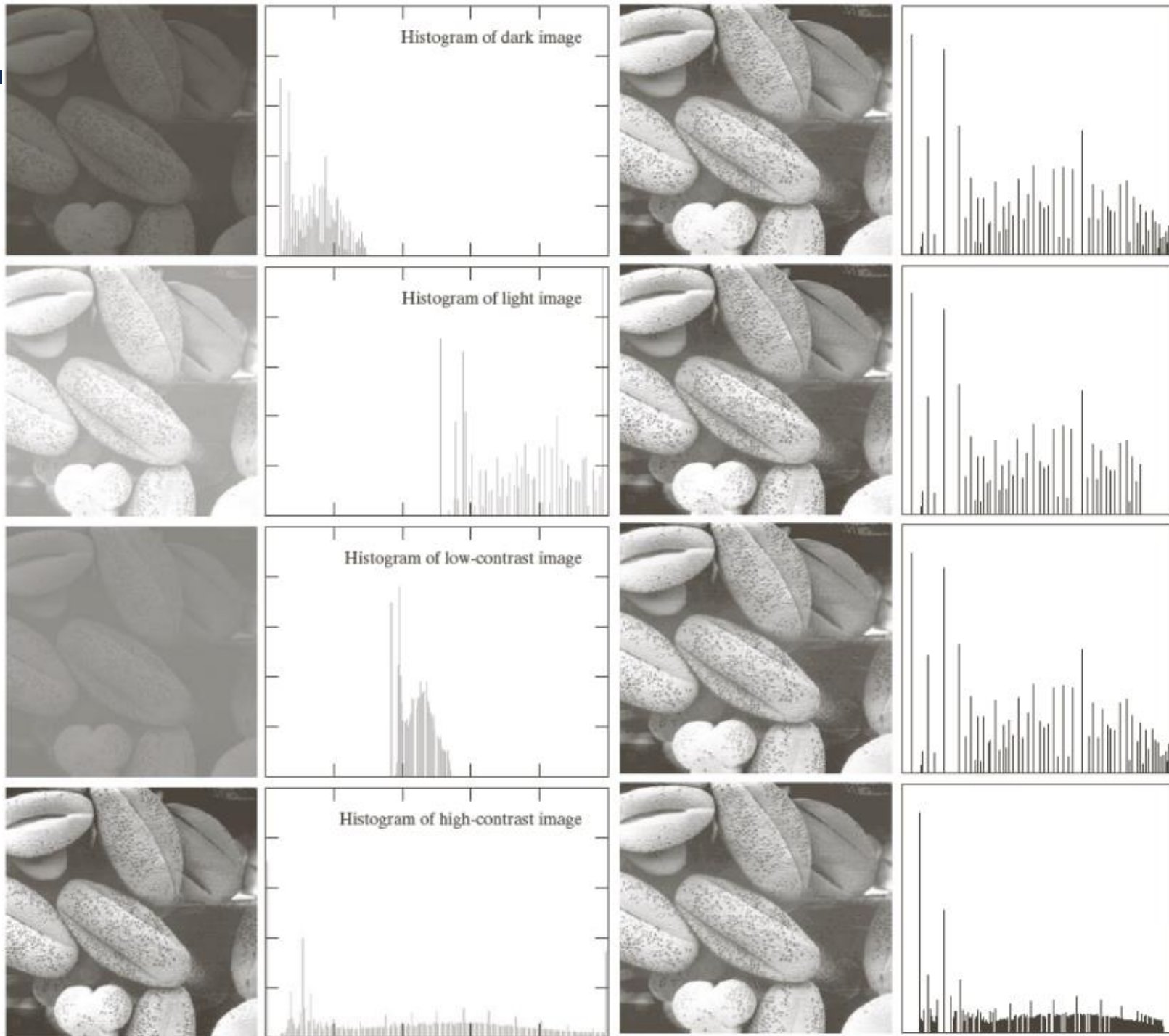
Gray level r_k	n_k	$P(r_k) = n_k/N$ (pdf)	s_k (CDF)	$S_k \times (L-1)$	Hist. equalized level
0	0	0	0	0	0
1	8	0.32	0.32	2.24	2
2	8	0.32	0.64	4.48	4
3	2	0.08	0.72	5.04	5
4	0	0	0.72	5.04	5
5	7	0.28	1	7	7
6	0	0	1	7	7
7	0	0	1	7	7

EXAMPLE CONT.

Grey Level	0	2	4	5	7
Frequency	0	8	8	2	7

2	4	2	2	2
4	7	5	7	4
4	7	7	7	4
4	7	5	7	4
2	2	2	4	2

Modified Image according to equalized histogram





SPATIAL FILTERING

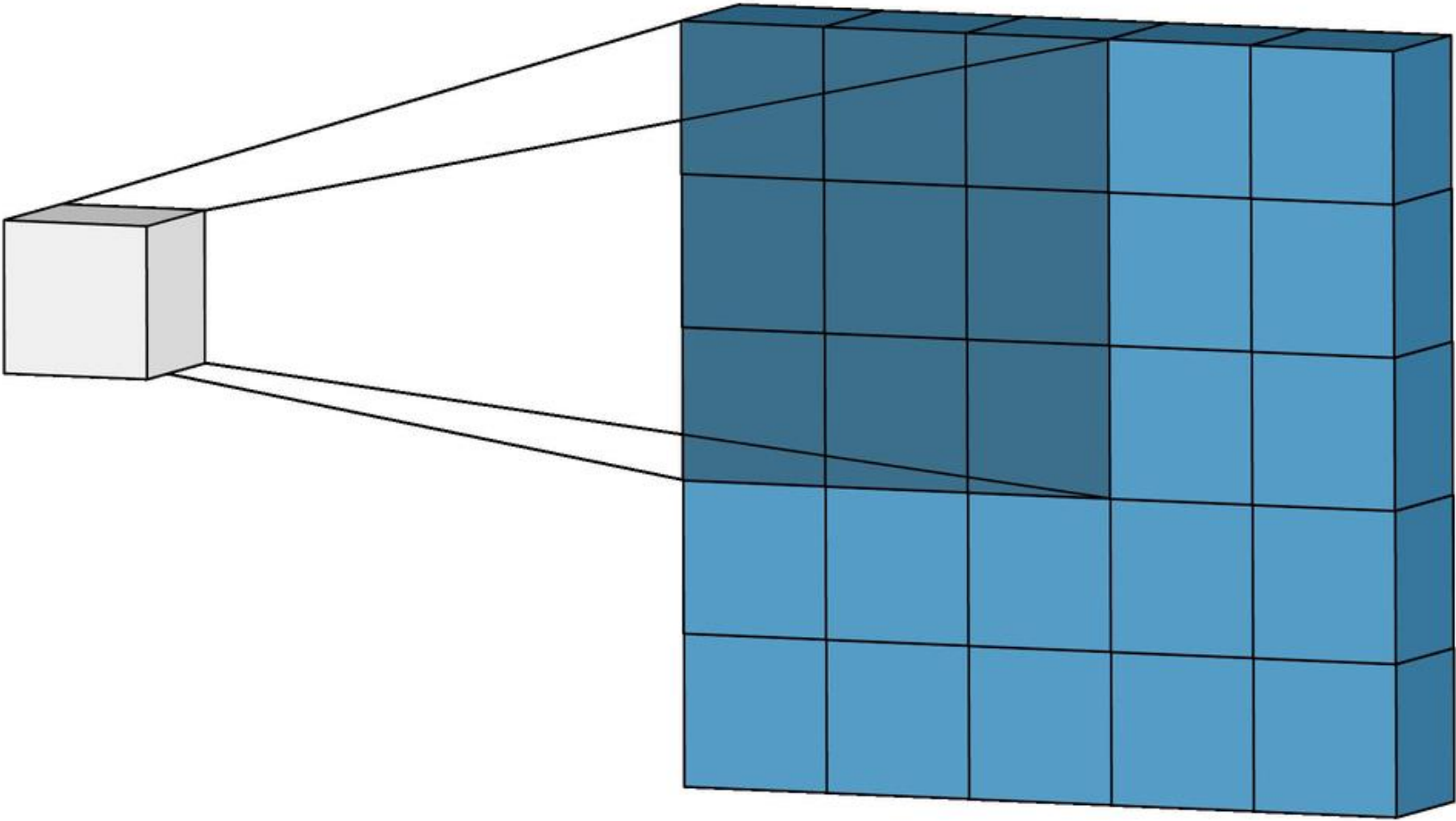


FILTER KERNEL

- The kernel is an array whose size defines the neighborhood of operation
- Whose coefficients determine the nature of the filter.
- Other terms used to refer to a spatial filter kernel are mask, template, and window.

LINEAR SPATIAL FILTERING

- The process consists simply of moving the filter mask from point to point in an image.
- At each point (x,y) the response of the filter at that point is calculated using a predefined relationship.
- A linear spatial filter performs a sum-of-products operation between an image f and a filter kernel, w .

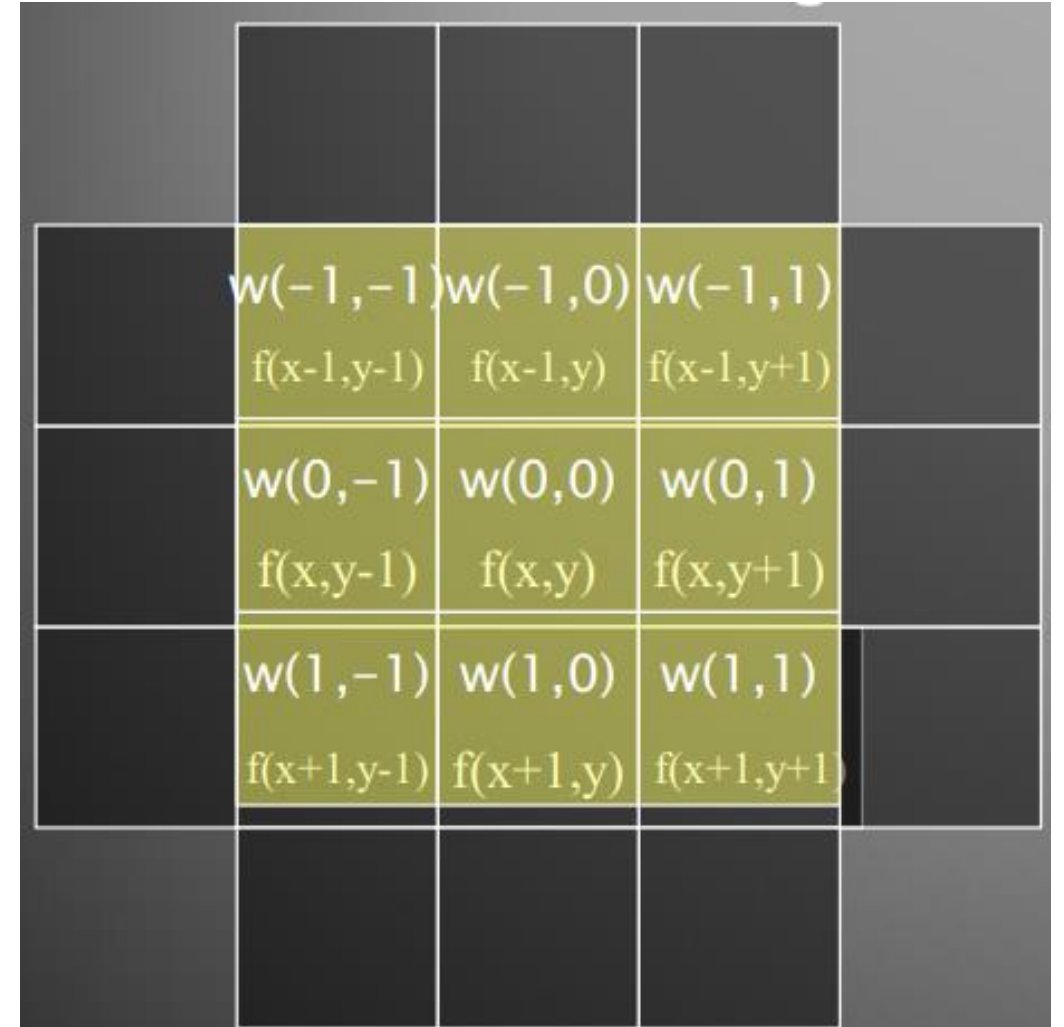


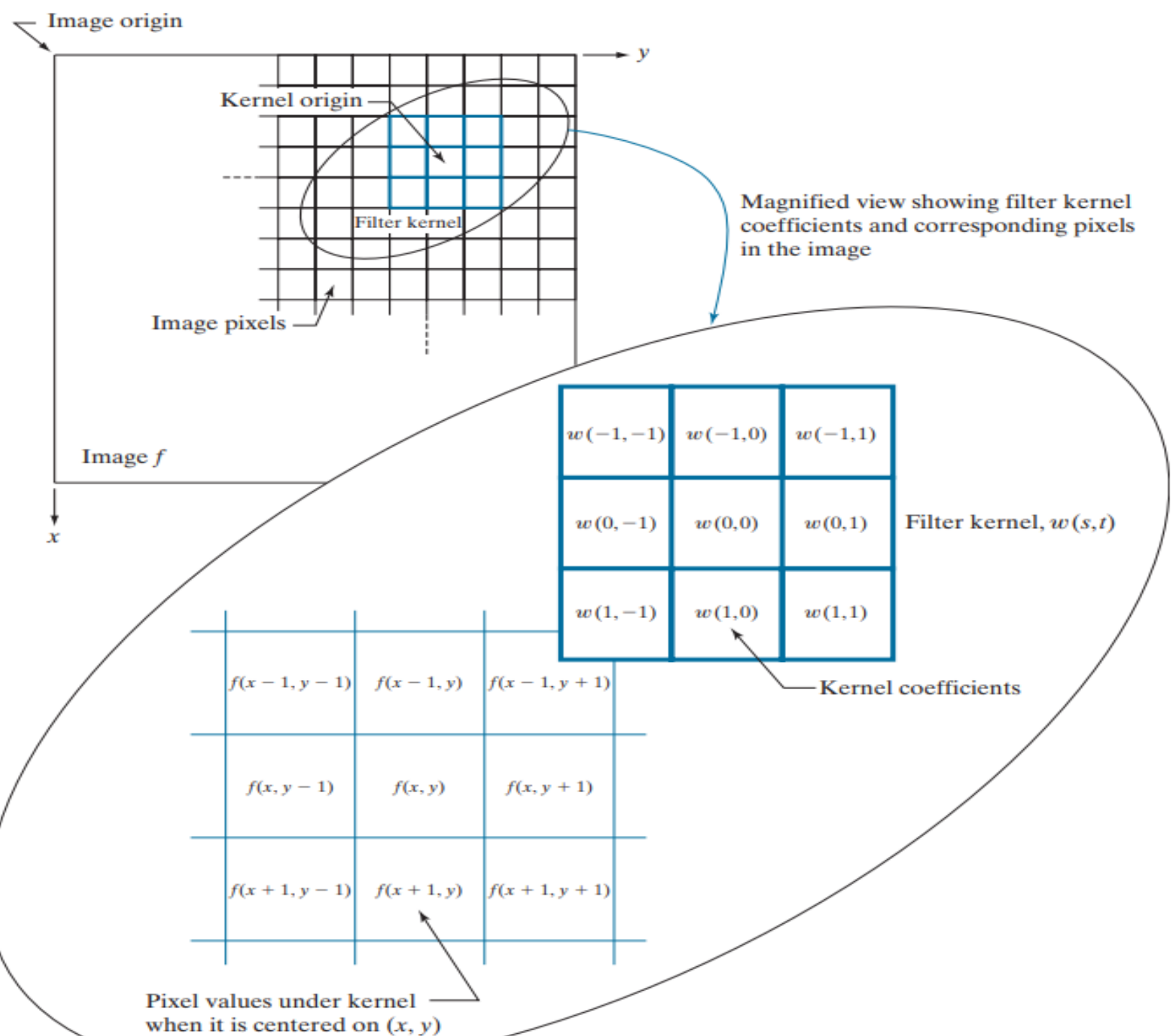
TRANSFORMED VALUE OF A PIXEL

- The result is the sum of products of the mask coefficients with the corresponding pixels directly under the mask.

$$g(x, y) = w(-1, -1)f(x - 1, y - 1) + w(-1, 0)f(x - 1, y) + \dots \\ + w(0, 0)f(x, y) + \dots + w(1, 1)f(x + 1, y + 1)$$

$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t)f(x + s, y + t)$$





Spatial Filtering (Neighborhood Processing)

Linear Spatial Filtering

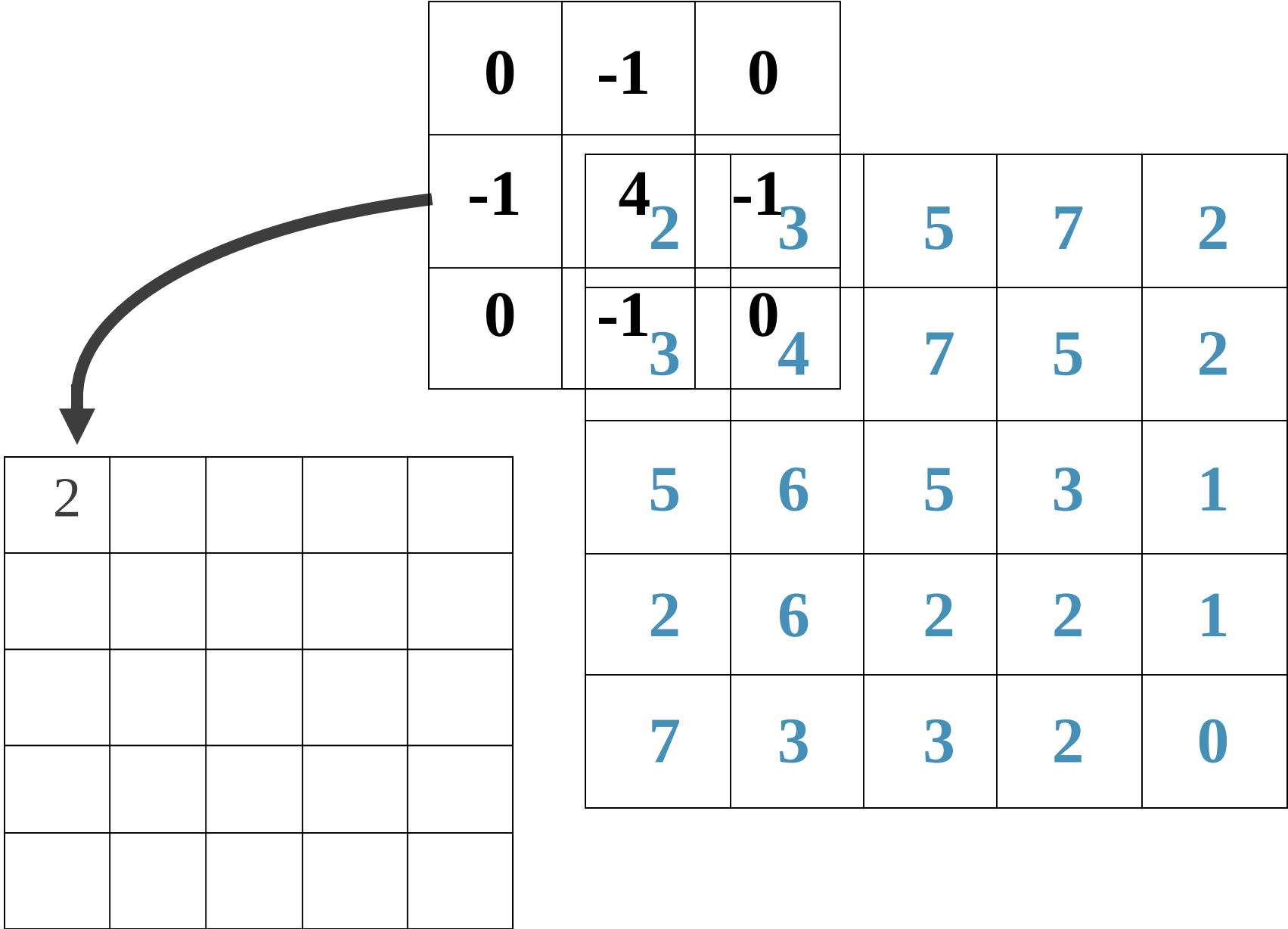
0	-1	0
-1	4	-1
0	-1	0

w: positioned so that its center
coefficient is coincident with the origin of f

2	3	5	7	2
3	4	7	5	2
5	6	5	3	1
2	6	2	2	1
7	3	3	2	0



0	-1	0			
-1	4 ₂	-1 ₃	5	7	2
0	-1 ₃	0 ₄	7	5	2
	5	6	5	3	1
	2	6	2	2	1
	7	3	3	2	0





2	1			

0	-1	0		
-1 ₂	4 ₃	-1 ₅	7	2
0 ₃	-1 ₄	0 ₇	5	2
5	6	5	3	1
2	6	2	2	1
7	3	3	2	0



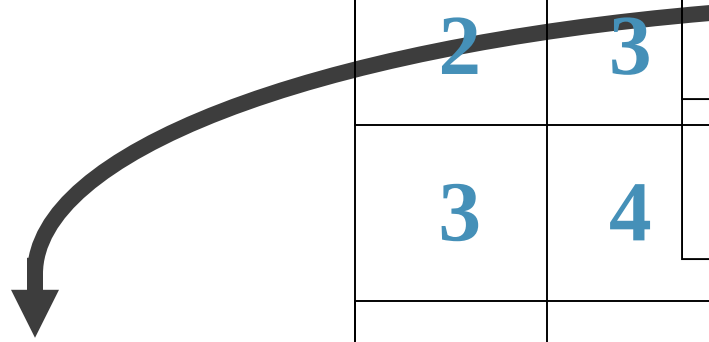
2	1	3		



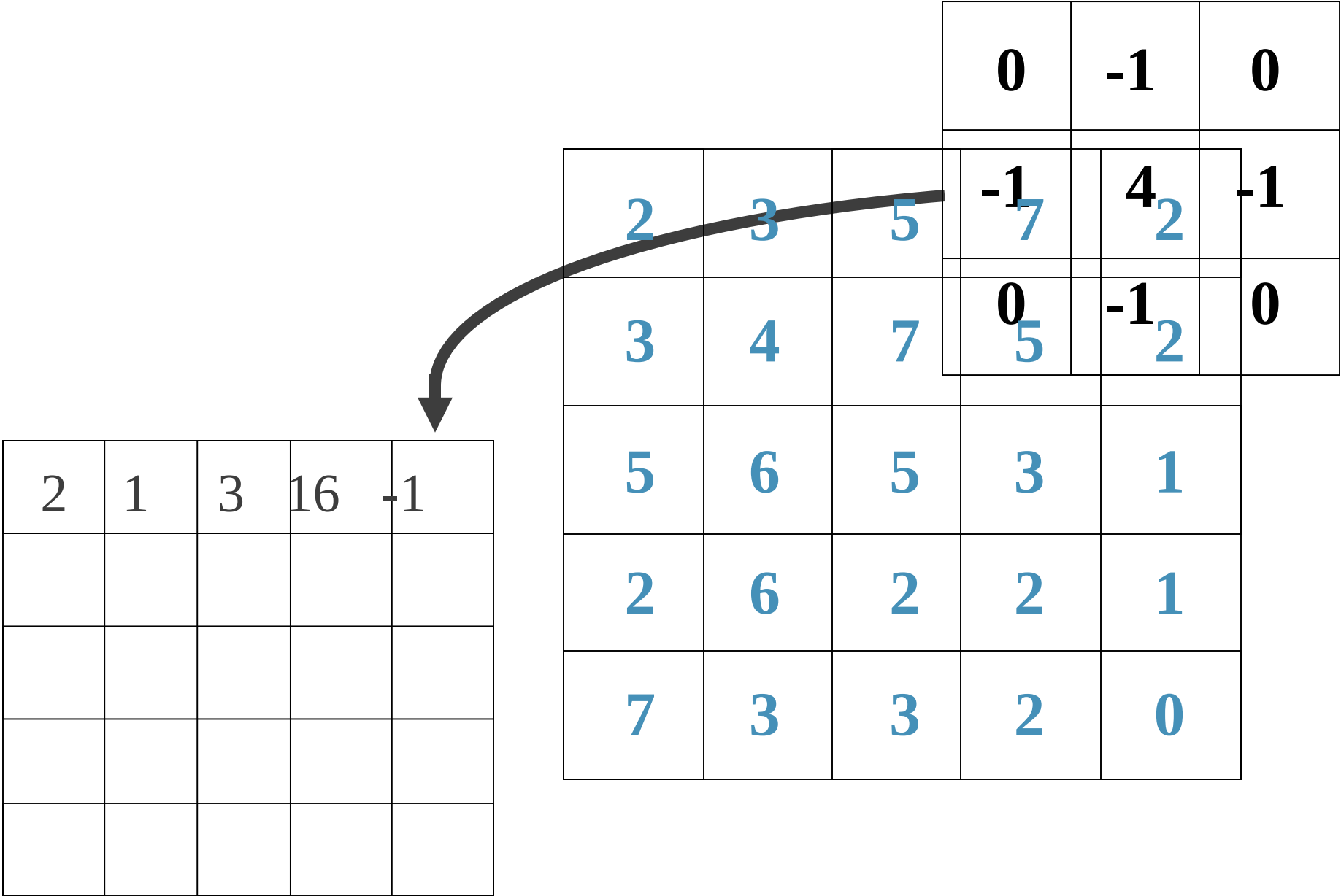
		0	-1	0	
2	-1	4	-1	2	
3	0	-1	0	2	
5	6	5	3	1	
2	6	2	2	1	
7	3	3	2	0	

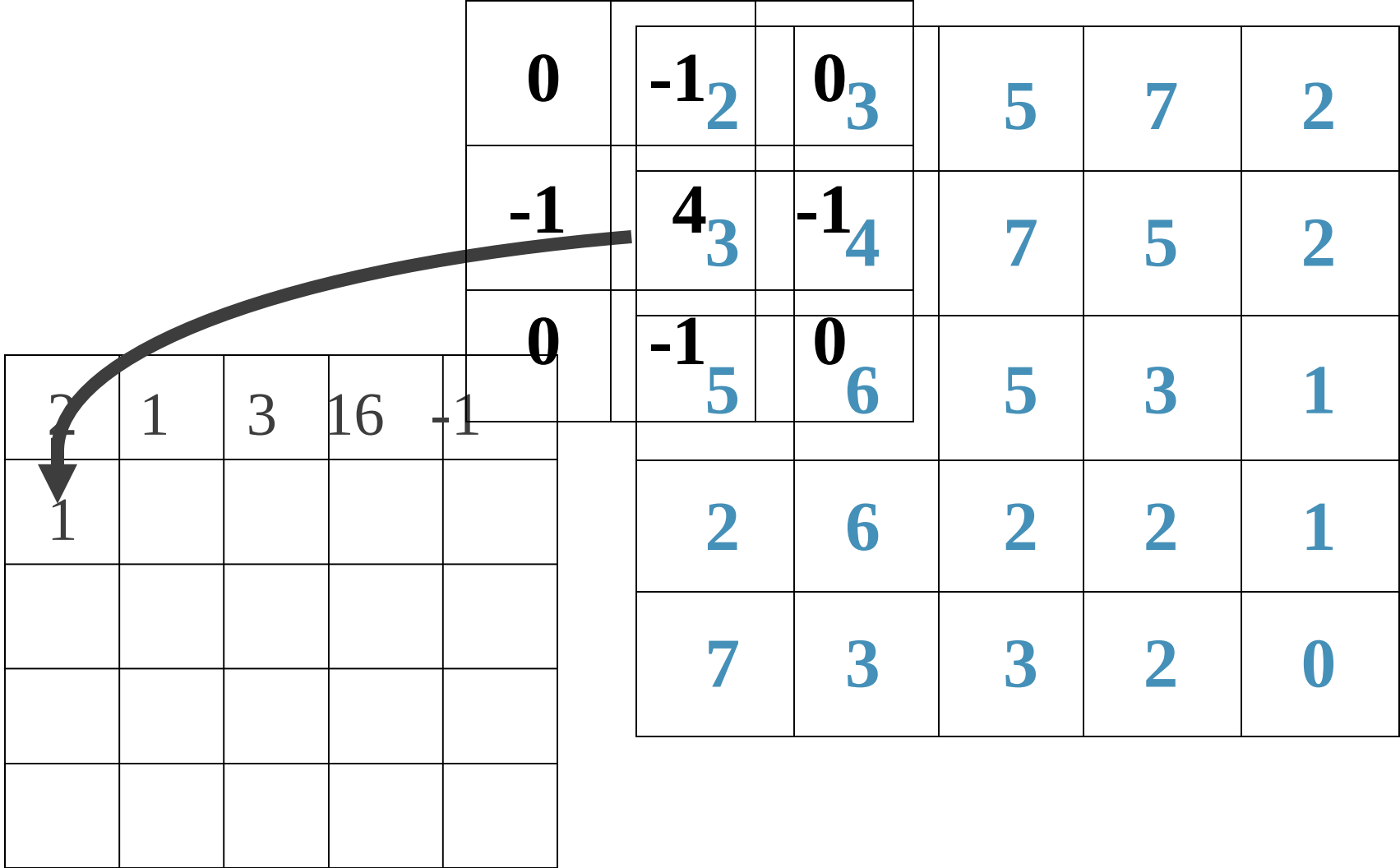


2	1	3	16	



		0		-1		0		
		-1		4		-1		
2		5		7		2		
3		0		-1		0		
4		7		5		2		
5		5		3		1		
6								
2		2		2		1		
6								
7		3		2		0		
3		3						







2	1	3	16	-1
1	-3			

